The Iberian setting: what the data holds, and what killed each design

Theodor Mello
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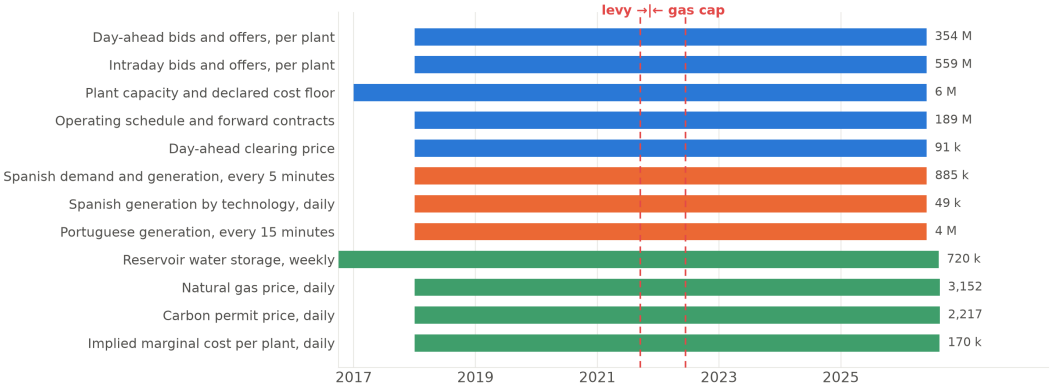
The decision on the table

Keep the Iberian setting, or kill it.

Three things to put to you:

1. **The data** is good — and it is built, audited and documented.
2. **One result** already holds up: market power roughly doubled while measured concentration *fell*.
3. **Twelve designs have died.** Not twelve accidents — three recurring causes, and they are properties of the setting.

What we hold: nine years, fully re-fetchable

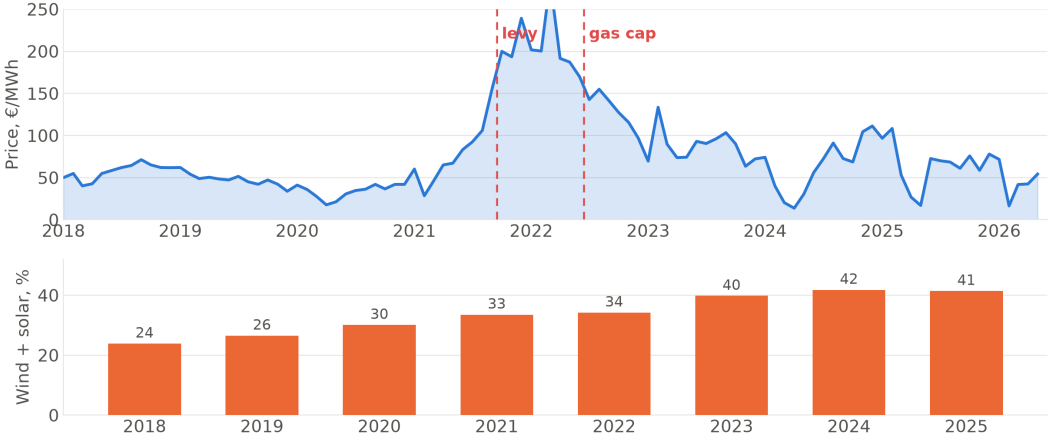


Both Spanish interventions sit **inside** the window: **3.7 years before the levy**, and **2.4 years after the cap ended**.

Why this data is unusual

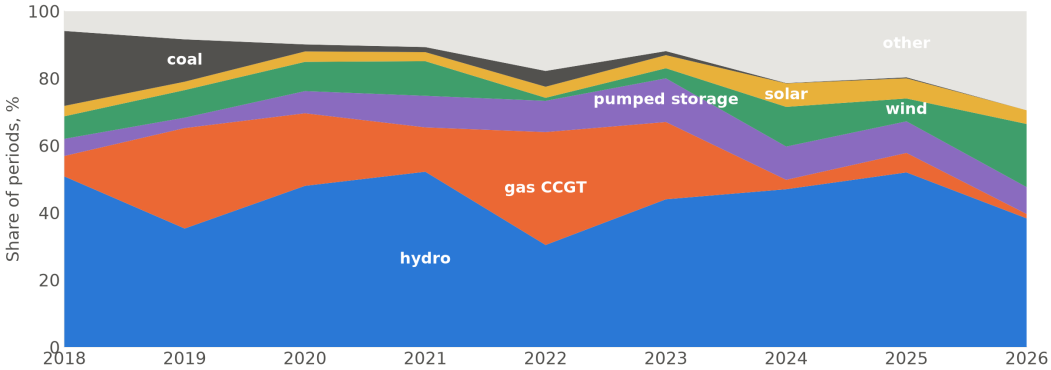
Full bid curves	354 M unit-level day-ahead offers — **accepted *and* rejected**. We observe firms' supply *functions*, not just realised supply.
Declared start-up costs	Every plant states, each day, the minimum revenue it needs to run — a fixed €/day and a variable €/MWh term.
Observable contract position	The volume sold forward, which blunts the incentive to withhold. Bilateral contracts run **39–49%** of scheduled sell volume.
Two countries, one auction	Spain and Portugal clear jointly against one merged bid stack — same price, same forward curve, different regulator.

The window: a price shock, and a VRE share that doubled



Day-ahead price 34 €/MWh (2020) → 168 (2022) → 46 (2026). Wind and solar 24% → 42% of Spanish generation.

Who sets the price



Hydro is the modal marginal technology in every year — 30–52% of periods. Gas is second and collapses after the cap. This is why every design here ended up about hydro.

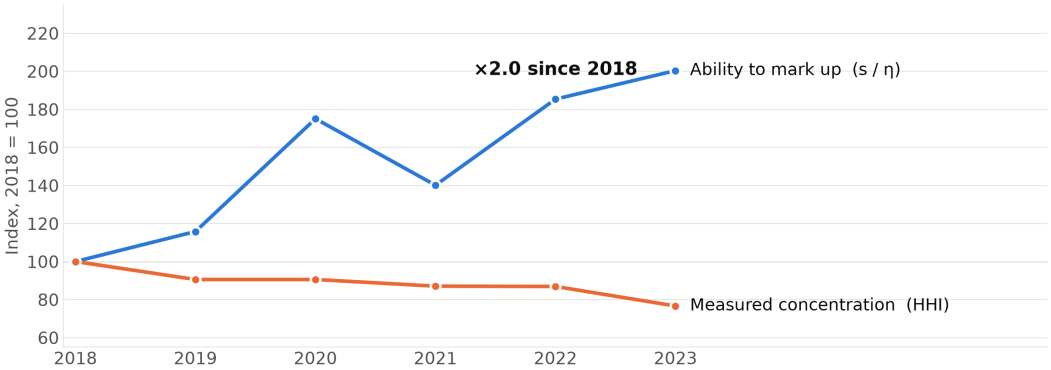
Who owns what: the mapping, and its ceiling

OMIE publishes no owner. Built in four layers from ESIOS, `cab` plant names and a hand-verified prefix dictionary.

what	coverage
Units in the registry (ES 4,960 / PT 150)	5,137
Near-marginal offered volume → market agent	**89.1%**
Generation volume → parent firm (68 firms)	82.4%
Dispatchable capacity → parent firm	**96.0%**
Generation volume bid by a *named plant* (technology readable)	47.7%

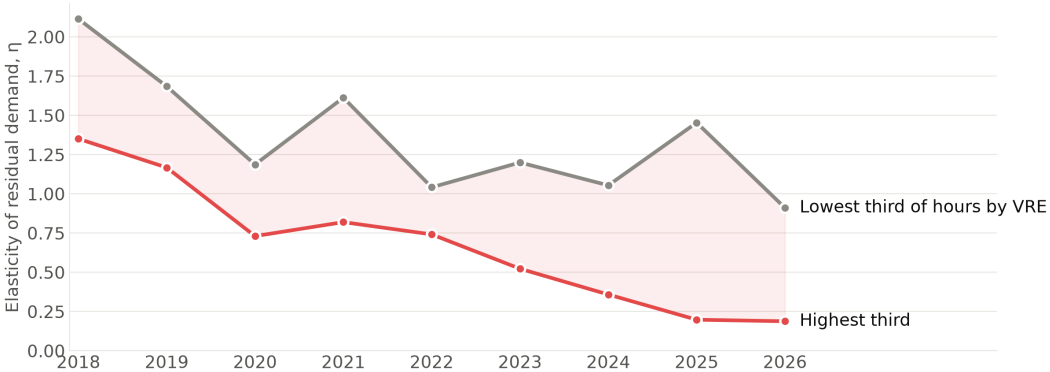
Why only 48%? The rest is **not unparsed — it is bid by portfolios, not plants**. `EDPSVD1` “VENTA RE” is 6,500 MW of mixed renewables in one unit: there is no single technology to name.

The result that holds: market power doubled as concentration fell



$L_i = s_i/\eta$. Shares barely moved; η **halved** — 1.83 (2018) → 0.85 (2023). For Iberdrola/Endesa the implied Lerner floor goes **11%** → **21%**, and to **34%** in the highest-VRE third of 2023 hours.

The mechanism: renewables thin the market *at the margin*



Wind and solar bid at zero and sit far inframarginal — they do not flatten the curve where the price is set, they push the crossing point into a region with fewer flexible units.

The diagnosis: three causes, not twelve accidents

1	**The treatment has no cross-section.** Levy, gas cap and renewable share hit every unit at once. Date fixed effects and the design are mutually exclusive.
2	**Any summary of "how a plant bid" changes sign depending on how it is built.** The median sell offer is €0 and the 99th percentile €1,000, so a few extreme offers decide the answer — and none of them can be read as withholding rather than a genuinely high cost.
3	**A hard cluster ceiling.** Only 32 parent firms own dispatchable plant. Any treatment varying at firm level has a few dozen clusters.
+3	**No usable comparison, or too few observations.** The single-date event study, the run-of-river control group and the 10 MW RDD each failed before estimation.

Everything that is a *direct computation* held: the elasticity decline, firm and technology coverage, contract shares, firm-specific residual demand, the implied-marginal-cost ordering by technology.

Everything built as a *constructed behavioural measure* collapsed.

Appendix: how the three numbers are built

A1. Residual demand, and its elasticity η

Firm i does not face market demand, it faces what is left after every rival has supplied. For period t:

$$RD_{it}(p) \;=\; D_t(p) \;-\; \sum_{j \neq i} S_{jt}(p)$$

We build the *market-wide* version, $RD_t(p) = D_t(p) - S_t(p)$, which is the curve facing an infinitesimally small participant. Its slope at the clearing price p_t^* is

$$|RD_t'(p_t^*)| \;=\; |D_t'(p_t^*)| \;+\; S_t'(p_t^*)$$

Submitted curves are **step functions and have no derivative**, so the slope is estimated as the density of offered volume in a price band of width $2h$ around the clearing price:

$$|RD_t'(p_t^*)| \;\approx\; \frac{\text{sell MW} + \text{buy MW within } p_t^* \pm h}{2h} \qquad \Longrightarrow \qquad \eta_t \;=\; \frac{|RD_t'|\, p_t^*}{Q_t}$$

η_t is scale-free, so it is comparable across years in which the price level moved by a factor of five.

A1. How η is applied to the data

Unit of observation	One market period (one settlement interval for the whole Iberian market)
Frequency	Hourly to 30 September 2025; **quarter-hourly** from 1 October 2025, when the market changed its settlement interval
Sample	**79,036 periods**, 2018–2026 — 62,721 hourly and 16,315 quarter-hourly
Input	Every *submitted* bid and offer, accepted **and** rejected (354 M rows). Cleared-only data cannot give a slope
Bandwidth h	**5 €/MWh** in the reported results; 1, 2 and 10 also computed, rank correlation 0.92–0.94
Units	$\lvert RD' \rvert$ in MW per €/MWh; p^* in €/MWh; Q in MWh; η dimensionless
Periods dropped	Market splitting (Spain and Portugal price apart); whole-day plant dropouts; empty price bands; $p^* \leq 1$ €/MWh, where η is not defined

A2. From η to the markup

Firm i picks its output taking the residual demand curve as given:

$$\max_{q_i} \; p(q_i)\, q_i - C_i(q_i) \qquad \Longrightarrow \qquad p + q_i \frac{\partial p}{\partial q_i} = C_i'(q_i)$$

and since $\partial p / \partial q_i = 1/RD_i'(p)$, the markup over marginal cost is

$$p - C_i' \;=\; \frac{q_i}{|RD_i'|} \qquad \Longrightarrow \qquad L_i \;\equiv\; \frac{p - C_i'}{p} \;=\; \frac{q_i}{|RD_i'|\, p} \;=\; \frac{s_i}{\eta}$$

with $s_i = q_i/Q$. Volume already sold forward at a fixed price earns nothing extra when p rises, so with a contract position θ_i the markup becomes $(q_i - \theta_i)/|RD_i'|$.

Why this is a lower bound. A firm's own supply is excluded from its own residual demand, so $|RD_i'| \le |RD'|$: every real firm faces a curve *at least* as steep as the market-wide one. Hence $L_i \ge s_i/\eta$ — the least markup consistent with profit maximisation, not an estimate of the markup taken.

A2. How the markup is applied to the data

η	Median across all periods within the year, from A1
s_i	Annual generation share by business group, published by the competition regulator, covering Spain and Portugal together, 2018–2023 — so the yearly figures pair an annual share with a per-period elasticity
Also computed	Share per firm *per period* from cleared volume, 1,101,980 firm-period observations, which avoids treating a share as constant across a year (within-year variation is large)
Units	L_i is a share of the price, dimensionless; the markup in €/MWh is $L_i \times p^*$
Reported	For the average of the two largest firms: L rises **11% (2018)** → **21% (2023)**, and to **34%** in the highest-renewable third of 2023 hours
Two biases, opposite	**Understated** on the elasticity side (the market-wide curve is flatter than any firm's own). **Overstated** on the share side: the published share counts volume sold bilaterally, which is not exposed to the day-ahead price

A3. Concentration, and why it moves the other way

The standard concentration measure is the sum of squared shares:

$$H \;=\; 10{,}000 \times \sum_i s_i^2$$

Source	Published by the competition regulator, annual, Spain and Portugal together, 2008–2023
Computed on	The *operating schedule* — day-ahead cleared volume **plus** bilateral contracts and self-scheduling. On January 2024 that is **1.73**× day-ahead cleared volume
Units	Index on 0–10,000; 1,000 is roughly ten equal firms

The two measures can move in opposite directions because they share only one of their two ingredients. H depends on shares alone. The Lerner index $L_i = s_i/\eta$ depends on shares *and* on the slope of residual demand. Between 2018 and 2023 shares barely moved — H fell 1,314 → 1,007 — while η fell 1.83 → 0.85. The same share converts into twice the markup.